Light

GCSE Computer Science - Programming Testing and Robustness (Set 2)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Programming Testing And Robustness

Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What does 'anticipating misuse' mean in defensive design?

- ○ **A.** Assuming all users will follow instructions perfectly
- ○ **B.** Considering how users might enter incorrect or malicious data and designing the program to handle it
- ○ **C.** Making the program interface more visually appealing
- ○ **D.** Removing all input options from the program

Q2 (1 mark)

What is erroneous test data?

- ○ **A.** Data that is within the valid range but causes unexpected results
- ○ **B.** Data that is at the boundary of the valid range
- ○ **C.** Data that the program should accept without any errors
- ○ **D.** Data of the wrong data type that should be rejected by the program

Q3 (1 mark)

What is invalid test data?

- ○ **A.** Data of the correct data type but outside the acceptable range that should be rejected
- ○ **B.** Data of the wrong data type entirely
- ○ **C.** Data that is accepted by the program and processed normally
- ○ **D.** Data that tests the exact limits of the acceptable range

Q4 (1 mark)

Which statement best describes what happens when a syntax error is present in code?

- ○ **A.** The program cannot be translated or run at all
- ○ **B.** The program runs more slowly than expected
- ○ **C.** The program runs but produces incorrect output
- ○ **D.** The program crashes only when certain inputs are provided

Q5 (1 mark)

Which statement best describes the behaviour of a program with a logic error?

- ○ **A.** The program will not run because the translator detects the error
- ○ **B.** The program displays an error message about incorrect syntax
- ○ **C.** The program cannot be saved to the computer
- ○ **D.** The program runs successfully but produces incorrect or unexpected results

Q6 (1 mark)

Why do programmers test their programs?

- ○ **A.** To identify and fix errors before users encounter them
- ○ **B.** To add new features to the program
- ○ **C.** To make the code run faster
- ○ **D.** To reduce the size of the program files

Q7 (1 mark)

What is a sub program in programming?

- ○ **A.** A program that runs in the background while another program is active

- ○ **B.** A section of code that performs a specific task and can be called from other parts of the program
- ○ **C.** A temporary copy of a program stored in RAM
- ○ **D.** A program that has fewer lines of code than the main program

Q8 (1 mark)

What is the purpose of password verification during account creation?

- ○ **A.** To ensure the user has typed their intended password correctly by entering it twice
- ○ **B.** To make the password harder to guess
- ○ **C.** To encrypt the password for secure storage
- ○ **D.** To check that the password is not already in use by another user

Q9 (1 mark)

Which variable name demonstrates poor naming conventions?

- ○ **A.** studentAge
- ○ **B.** numberOfItems
- ○ **C.** x1
- ○ **D.** totalPrice

Q10 (1 mark)

When does final (terminal) testing typically occur?

- ○ **A.** Before any code has been written
- ○ **B.** Continuously throughout the development process
- ○ **C.** Only when users report problems with the software
- ○ **D.** At the end of development when the complete program is ready

Q11 (1 mark)

What is an advantage of iterative testing during development?

- ○ **A.** It reduces the total amount of testing needed
- ○ **B.** Errors are found early when they are easier and cheaper to fix
- ○ **C.** It eliminates the need for final testing
- ○ **D.** It automatically corrects any errors that are found

Q12 (1 mark)

What type of information should comments in code typically explain?

- ○ **A.** Every single line of code, regardless of complexity
- ○ **B.** The names of all variables used in the program
- ○ **C.** Personal opinions about whether the code is good or bad

○ **D.** The purpose and reasoning behind code sections, especially complex logic

Q13 (2 marks)

A cinema booking system requires users to enter the number of tickets (between 1 and 10).
Which validation check is most appropriate?

- ○ **A.** Length check
- ○ **B.** Format check
- ○ **C.** Presence check
- ○ **D.** Range check

Q14 (2 marks)

A student writes the following Python code:

```
for i in range(10)
    print(i)
```

The code will not run. What type of error is present and what is the problem?

- ○ **A.** Logic error - the loop should start from 1 instead of 0
- ○ **B.** Runtime error - the range function is not defined
- ○ **C.** Logic error - the variable i should be called counter
- ○ **D.** Syntax error - the colon is missing after range(10)

Q15 (2 marks)

A program calculates a 10% discount on a price:

```
discountedPrice = price + (price * 0.10)
```

The program runs but customers complain they are being charged more, not less.
What type of error is this?

- ○ **A.** Logic error - the code adds the discount instead of subtracting it
- ○ **B.** Syntax error - the brackets are incorrect
- ○ **C.** Runtime error - the program crashed during calculation
- ○ **D.** Syntax error - the variable names are too long

Q16 (2 marks)

A program accepts test scores from 0 to 100 inclusive.
Which of the following represents appropriate boundary test data?

- ○ **A.** 50 and 75
- ○ **B.** -1, 0, 100, and 101
- ○ **C.** 0 and 100 only
- ○ **D.** 25, 50, and 75

Q17 (2 marks)

A company wants to improve security beyond just username and password.
Which method would provide two-factor authentication?

- ○ **A.** Requiring a password and a code sent to the user's phone
- ○ **B.** Requiring the user to answer two security questions

- ○ **C.** Requiring a longer password with special characters
- ○ **D.** Requiring the user to enter the password twice

Q18 (2 marks)

A program asks users to enter their date of birth.
Which defensive design technique would prevent users entering future dates?

- ○ **A.** Presence check
- ○ **B.** Range check comparing the date against today's date
- ○ **C.** Type check to ensure it is a date format
- ○ **D.** Length check on the date string

Q19 (2 marks)

Why is consistent indentation particularly important in Python programs?

- ○ **A.** Python uses indentation to define code blocks, so incorrect indentation causes errors
- ○ **B.** It makes the program run faster
- ○ **C.** It reduces the file size of the program
- ○ **D.** It allows the program to use less memory

Q20 (2 marks)

A test plan document records the results of testing.

Which combination of information should each test in the plan include?

- ○ **A.** Programmer name, date, and time only
- ○ **B.** Test number, test data, expected result, actual result, and pass/fail status
- ○ **C.** File name and file size only
- ○ **D.** Number of lines of code tested

Q21 (2 marks)

A program requires users to enter an email address.
Which validation check would verify the email contains an @ symbol?

- ○ **A.** Range check
- ○ **B.** Length check
- ○ **C.** Format check
- ○ **D.** Presence check

Q22 (2 marks)

A programmer is struggling to find an error in their code.
Which type of error is typically harder to locate and why?

- ○ **A.** Syntax errors, because they produce no error messages
- ○ **B.** Logic errors, because they prevent the program from running
- ○ **C.** Syntax errors, because they only appear when certain inputs are used

○ **D.** Logic errors, because the program runs without error messages but produces wrong output

Q23 (2 marks)

A programmer splits their code into several sub programs (functions and procedures).
How does this improve code maintenance?

○ **A.** It makes the program use less memory

○ **B.** It automatically prevents all errors from occurring

○ **C.** Each sub program can be understood, tested, and modified independently

○ **D.** It removes the need for comments in the code

Q24 (3 marks)

A web application allows users to search for products by entering a search term.
The programmer builds the database query by directly inserting the user's input into the SQL statement.
Analyse why this is a security risk and what defensive measure should be used.

○ **A.** This is a security risk because SQL injection attacks can manipulate the query to access, modify, or delete unauthorised data; the programmer should use parameterised queries that separate user input from the SQL command structure

○ **B.** This is only risky if the database contains financial data; otherwise direct insertion is acceptable

○ **C.** This is safe because databases automatically filter dangerous input

○ **D.** The only risk is slow performance; security is not affected by how queries are built

Q25 (3 marks)

A program validates that a username is between 3 and 15 characters long.
Analyse which test data set would provide the most thorough testing of this validation.

○ **A.** Normal: 'user1', 'testname'; Boundary: 'abc', 'abcdefghijklmno'; Invalid: 'ab', 'abcdefghijklmnop'; Erroneous: 12345 (integer)

○ **B.** Only testing with 'admin' and 'user'

○ **C.** Testing with 100 different valid usernames

○ **D.** Only testing with empty strings

Q26 (3 marks)

A program should calculate the total cost including 20% VAT:

```
basePrice = 100
vatRate = 20
totalPrice = basePrice + vatRate
```

The program outputs 120, but 100 items at this price are totalling incorrectly in reports.
Analyse the error and explain the correct calculation.

○ **A.** Syntax error - vatRate should be in quotes

○ **B.** Runtime error - the variables are not initialised

○ **C.** Logic error - the code adds 20 as a fixed amount instead of calculating 20% of the price; it should be basePrice + (basePrice * vatRate / 100) or basePrice * 1.20

○ **D.** No error exists - 120 is the correct result

Q27 (3 marks)

A developer is designing an authentication system for a new application. Analyse which combination of security measures would provide the most robust password protection.

○ **A.** Store passwords in plain text for easy comparison and allow unlimited login attempts

○ **B.** Hash passwords with a secure algorithm, add unique salt to each password, limit failed login attempts, and require minimum password complexity

○ **C.** Encrypt passwords with a reversible algorithm and display password hints publicly

○ **D.** Store passwords in a separate file on the server with no additional protection

Q28 (3 marks)

A programmer inherits code from a colleague who has left the company. The code has no comments, uses variable names like 'a', 'b', 'x1', 'temp2', and consists of one 500-line function.
Analyse how these issues affect maintainability and what improvements should be made.

○ **A.** The code is fine because it runs correctly; maintainability does not matter

○ **B.** Only the variable names need changing; long functions are acceptable

○ **C.** The lack of comments, meaningless variable names, and monolithic structure make the code extremely difficult to understand, modify, and debug; improvements should include adding explanatory

comments, using descriptive variable names, and breaking the code into smaller sub programs with single responsibilities

○ **D.** The code should be deleted and rewritten from scratch without examining what it does

Q29 (3 marks)

A software development team must decide between testing only at the end of the project versus testing throughout development.
Analyse the advantages of combining iterative and final testing approaches.

○ **A.** Testing only at the end is better because it saves time during development

○ **B.** Iterative testing alone is sufficient; final testing adds no value

○ **C.** Iterative testing finds bugs early when they are cheaper to fix and prevents bug accumulation, while final testing verifies the complete integrated system works correctly; combining both provides thorough quality assurance

○ **D.** Neither approach is necessary if the programmers are experienced

Q30 (3 marks)

An online banking system requires users to enter a sort code (format: XX-XX-XX where X is a digit) and an account number (8 digits).
Analyse which combination of validation checks would provide the most thorough input validation.

○ **A.** Presence check only - just ensure both fields are filled in

○ **B.** For sort code: presence check, length check (8 characters including hyphens), and format check (digit-digit-hyphen pattern); For

account number: presence check, length check (exactly 8 characters), type check (all digits), and possibly check digit validation

- ○ **C.** Format check only on both fields
- ○ **D.** No validation needed since users can see what they are typing

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.